



আন্তর্জাতিক ইসলামি বিশ্ববিদ্যালয় চট্টগ্রাম
الجامعة الإسلامية العالمية شيتاغونغ
International Islamic University Chittagong

LAB REPORT

EXPERIMENT NO : 00
EXPERIMENT NAME : Implementation of ER Diagram and SQL
Queries for a Student Enrollment
Database
COURSE CODE : CSE-2423
COURSE TITLE : Database Management Systems

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REMARK:

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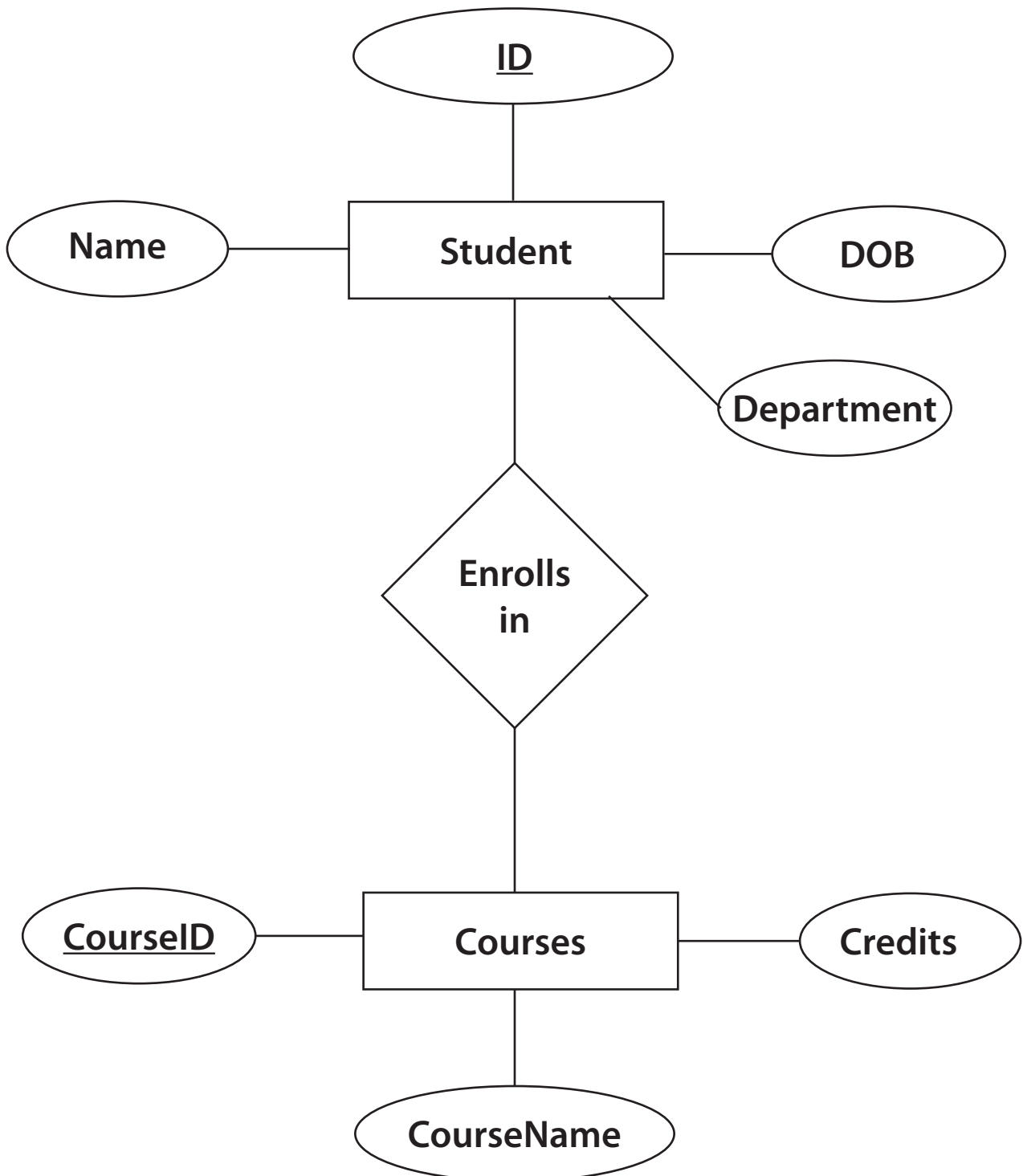
ER Diagram Task

Draw a simple ER Diagram based on these requirements:

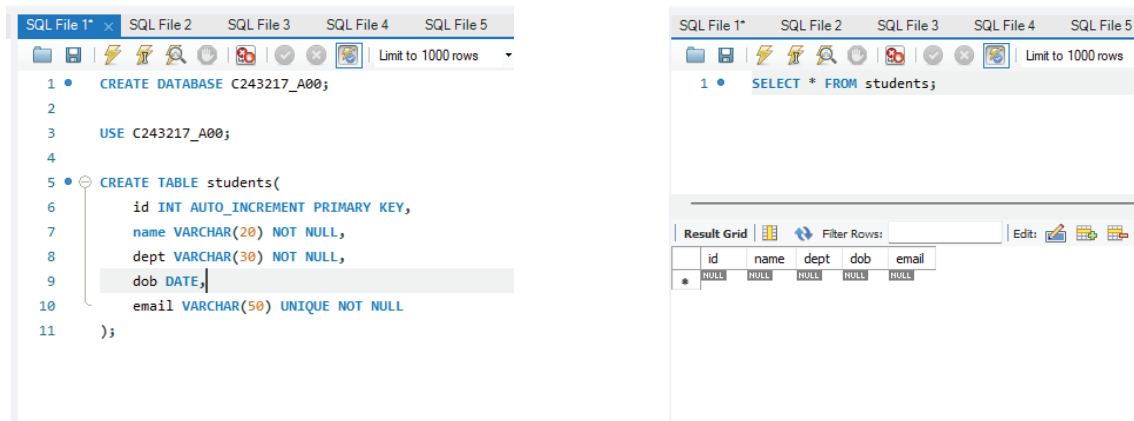
Entity 1: Student (Attributes: ID, Name, Department, DOB)

Entity 2: Course (Attributes: CourseID, CourseName, Credits)

Relationship: A Student enrolls in a Course.



1. ALTER the Students table to add a column named Email (Varchar 50):



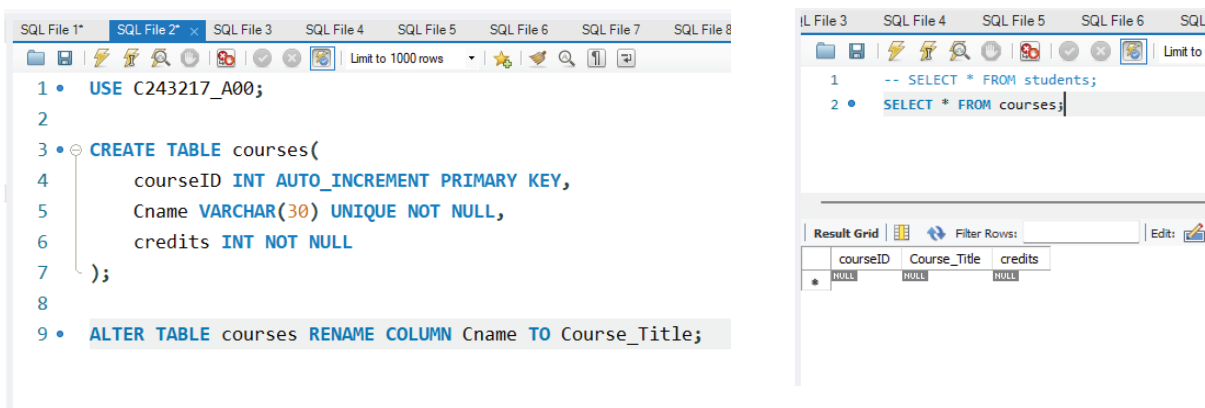
The screenshot shows two windows of an SQL IDE. The left window displays the SQL script for creating a database, using it, and creating a table named 'students'. The table has columns: id (INT AUTO_INCREMENT PRIMARY KEY), name (VARCHAR(20) NOT NULL), dept (VARCHAR(30) NOT NULL), dob (DATE), and email (VARCHAR(50) UNIQUE NOT NULL). The right window shows a query to select all data from the 'students' table, and below it, a 'Result Grid' showing a single row with all NULL values.

```
SQL File 1* x SQL File 2 SQL File 3 SQL File 4 SQL File 5
1 • CREATE DATABASE C243217_A00;
2
3 • USE C243217_A00;
4
5 • CREATE TABLE students(
6     id INT AUTO_INCREMENT PRIMARY KEY,
7     name VARCHAR(20) NOT NULL,
8     dept VARCHAR(30) NOT NULL,
9     dob DATE,
10    email VARCHAR(50) UNIQUE NOT NULL
11 );
```

```
SQL File 1* SQL File 2 SQL File 3 SQL File 4 SQL File 5
1 • SELECT * FROM students;
```

id	name	dept	dob	email
NULL	NULL	NULL	NULL	NULL

2. ALTER the Courses table to change the column name CName to Course_Title:



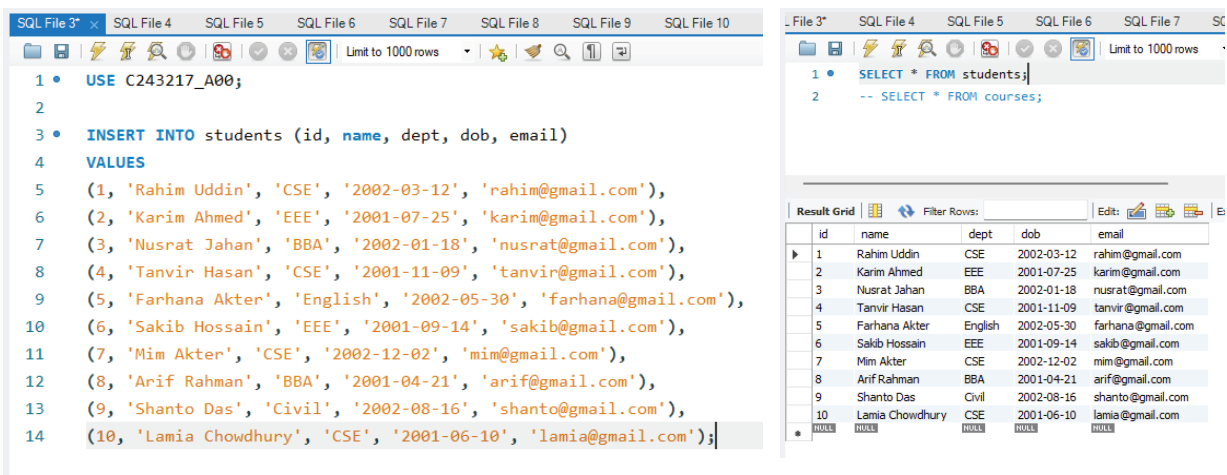
The screenshot shows two windows of an SQL IDE. The left window displays the SQL script for using the database, creating a table named 'courses', and then altering it to rename the 'Cname' column to 'Course_Title'. The right window shows a query to select all data from the 'courses' table, and below it, a 'Result Grid' showing a single row with all NULL values.

```
SQL File 1* SQL File 2* x SQL File 3 SQL File 4 SQL File 5 SQL File 6 SQL File 7 SQL File 8
1 • USE C243217_A00;
2
3 • CREATE TABLE courses(
4     courseID INT AUTO_INCREMENT PRIMARY KEY,
5     Cname VARCHAR(30) UNIQUE NOT NULL,
6     credits INT NOT NULL
7 );
8
9 • ALTER TABLE courses RENAME COLUMN Cname TO Course_Title;
```

```
SQL File 3 SQL File 4 SQL File 5 SQL File 6 SQL
1 -- SELECT * FROM students;
2 • SELECT * FROM courses;
```

courseID	Course_Title	credits
NULL	NULL	NULL

3. INSERT 10 students into the Students table:



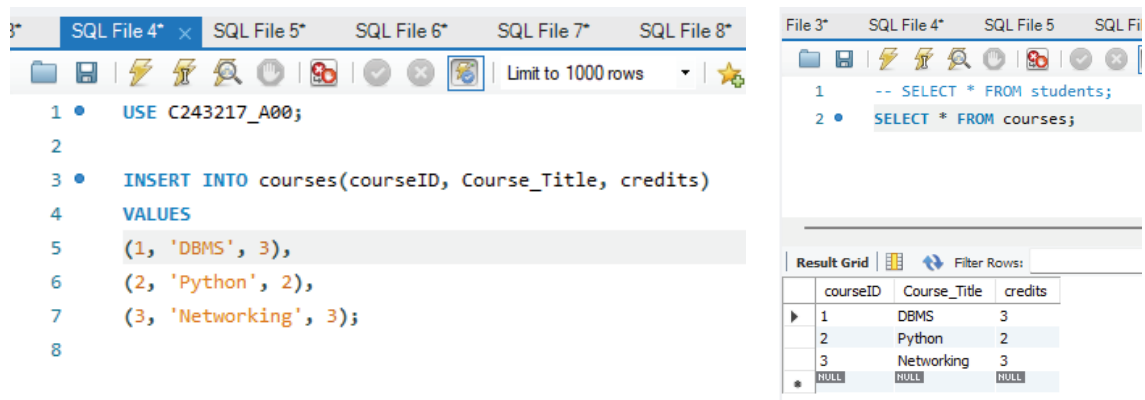
The screenshot shows two windows of an SQL IDE. The left window displays the SQL script for using the database and inserting 10 student records into the 'students' table. The right window shows a query to select all data from the 'students' table, and below it, a 'Result Grid' showing 10 rows of student data.

```
SQL File 3* x SQL File 4 SQL File 5 SQL File 6 SQL File 7 SQL File 8 SQL File 9 SQL File 10
1 • USE C243217_A00;
2
3 • INSERT INTO students (id, name, dept, dob, email)
4 VALUES
5 (1, 'Rahim Uddin', 'CSE', '2002-03-12', 'rahim@gmail.com'),
6 (2, 'Karim Ahmed', 'EEE', '2001-07-25', 'karim@gmail.com'),
7 (3, 'Nusrat Jahan', 'BBA', '2002-01-18', 'nusrat@gmail.com'),
8 (4, 'Tanvir Hasan', 'CSE', '2001-11-09', 'tanvir@gmail.com'),
9 (5, 'Farhana Akter', 'English', '2002-05-30', 'farhana@gmail.com'),
10 (6, 'Sakib Hossain', 'EEE', '2001-09-14', 'sakib@gmail.com'),
11 (7, 'Mim Akter', 'CSE', '2002-12-02', 'mim@gmail.com'),
12 (8, 'Arif Rahman', 'BBA', '2001-04-21', 'arif@gmail.com'),
13 (9, 'Shanto Das', 'Civil', '2002-08-16', 'shanto@gmail.com'),
14 (10, 'Lamia Chowdhury', 'CSE', '2001-06-10', 'lamia@gmail.com');
```

```
SQL File 3* SQL File 4 SQL File 5 SQL File 6 SQL File 7 SQL
1 • SELECT * FROM students;
2 -- SELECT * FROM courses;
```

id	name	dept	dob	email
1	Rahim Uddin	CSE	2002-03-12	rahim@gmail.com
2	Karim Ahmed	EEE	2001-07-25	karim@gmail.com
3	Nusrat Jahan	BBA	2002-01-18	nusrat@gmail.com
4	Tanvir Hasan	CSE	2001-11-09	tanvir@gmail.com
5	Farhana Akter	English	2002-05-30	farhana@gmail.com
6	Sakib Hossain	EEE	2001-09-14	sakib@gmail.com
7	Mim Akter	CSE	2002-12-02	mim@gmail.com
8	Arif Rahman	BBA	2001-04-21	arif@gmail.com
9	Shanto Das	Civil	2002-08-16	shanto@gmail.com
10	Lamia Chowdhury	CSE	2001-06-10	lamia@gmail.com

4. INSERT 3 courses into the Courses table (e.g., DBMS, Python, Networking):

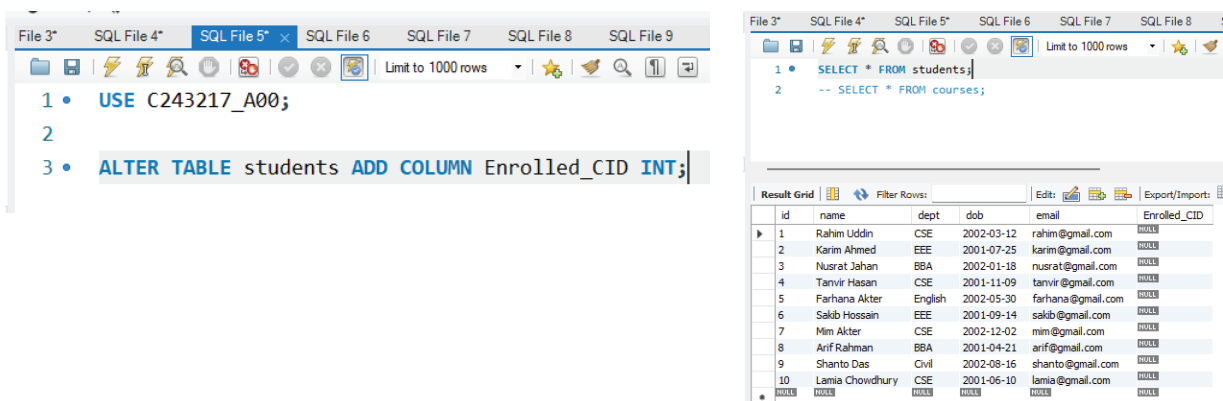


```
1 • USE C243217_A00;
2
3 • INSERT INTO courses(courseID, Course_Title, credits)
4 VALUES
5 (1, 'DBMS', 3),
6 (2, 'Python', 2),
7 (3, 'Networking', 3);
8
```

```
1 -- SELECT * FROM students;
2 • SELECT * FROM courses;
```

courseID	Course_Title	credits
1	DBMS	3
2	Python	2
3	Networking	3
NULL	NULL	NULL

5. The Manual Link: Use ALTER to add a column Enrolled_CID to the Students table:

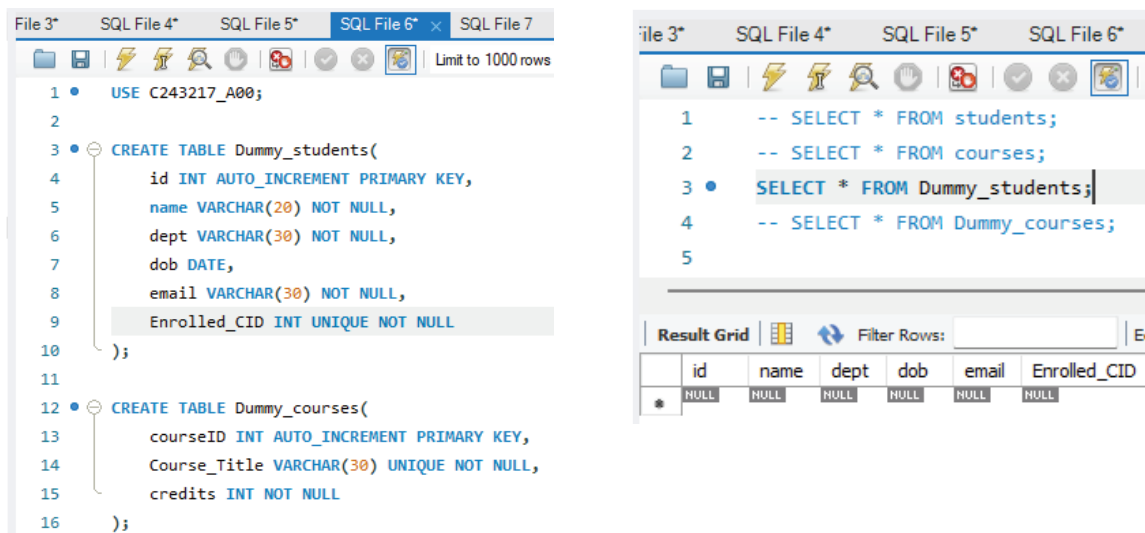


```
1 • USE C243217_A00;
2
3 • ALTER TABLE students ADD COLUMN Enrolled_CID INT;
```

```
1 SELECT * FROM students;
2 -- SELECT * FROM courses;
```

id	name	dept	dob	email	Enrolled_CID
1	Rahim Uddin	CSE	2002-03-12	rahim@gmail.com	NULL
2	Karim Ahmed	EEE	2001-07-25	karim@gmail.com	NULL
3	Nusrat Jahan	BBA	2002-01-18	nusrat@gmail.com	NULL
4	Tanvir Hasan	CSE	2001-11-09	tanvir@gmail.com	NULL
5	Farhana Akter	English	2002-05-30	farhana@gmail.com	NULL
6	Sakib Hossain	EEE	2001-09-14	sakib@gmail.com	NULL
7	Mim Akter	CSE	2002-12-02	mim@gmail.com	NULL
8	Arif Rahman	BBA	2001-04-21	arif@gmail.com	NULL
9	Shanto Das	Civil	2002-08-16	shanto@gmail.com	NULL
10	Lamia Chowdhury	CSE	2001-06-10	lamia@gmail.com	NULL

6. Create dummy tables. Apply the following on these dummy tables:



```
1 • USE C243217_A00;
2
3 • CREATE TABLE Dummy_students(
4     id INT AUTO_INCREMENT PRIMARY KEY,
5     name VARCHAR(20) NOT NULL,
6     dept VARCHAR(30) NOT NULL,
7     dob DATE,
8     email VARCHAR(30) NOT NULL,
9     Enrolled_CID INT UNIQUE NOT NULL
10 );
11
12 • CREATE TABLE Dummy_courses(
13     courseID INT AUTO_INCREMENT PRIMARY KEY,
14     Course_Title VARCHAR(30) UNIQUE NOT NULL,
15     credits INT NOT NULL
16 );
```

```
1 -- SELECT * FROM students;
2 -- SELECT * FROM courses;
3 • SELECT * FROM Dummy_students;
4 -- SELECT * FROM Dummy_courses;
5
```

id	name	dept	dob	email	Enrolled_CID
NULL	NULL	NULL	NULL	NULL	NULL

7. UPDATE/INSERT: Assign a CourseID to each student in that new column:

```
1 • USE C243217_A00;
2
3 • INSERT INTO Dummy_students (id, name, dept, dob, email, Enrolled_CID)
4 VALUES
5 (1, 'Rahim Uddin', 'CSE', '2002-03-12', 'rahim@gmail.com', 101),
6 (2, 'Karim Ahmed', 'EEE', '2001-07-25', 'karim@gmail.com', 102),
7 (3, 'Nusrat Jahan', 'BBA', '2002-01-18', 'nusrat@gmail.com', 103),
8 (4, 'Tanvir Hasan', 'CSE', '2001-11-09', 'tanvir@gmail.com', 104),
9 (5, 'Farhana Akter', 'ENG', '2002-05-30', 'farhana@gmail.com', 105),
10 (6, 'Sakib Hossain', 'EEE', '2001-09-14', 'sakib@gmail.com', 106),
11 (7, 'Mim Akter', 'CSE', '2002-12-02', 'mim@gmail.com', 107),
12 (8, 'Arif Rahman', 'BBA', '2001-04-21', 'arif@gmail.com', 108),
13 (9, 'Shanto Das', 'CIVIL', '2002-08-16', 'shanto@gmail.com', 109),
14 (10, 'Lamia Chowdhury', 'CSE', '2001-06-10', 'lamia@gmail.com', 110);
```

SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8

```
1 -- SELECT * FROM students;
2 -- SELECT * FROM courses;
3 • SELECT * FROM Dummy_students;
4 -- SELECT * FROM Dummy_courses;
5
```

Result Grid Filter Rows: Edit: Export/Import:

id	name	dept	dob	email	Enrolled_CID
1	Rahim Uddin	CSE	2002-03-12	rahim@gmail.com	101
2	Karim Ahmed	EEE	2001-07-25	karim@gmail.com	102
3	Nusrat Jahan	BBA	2002-01-18	nusrat@gmail.com	103
4	Tanvir Hasan	CSE	2001-11-09	tanvir@gmail.com	104
5	Farhana Akter	ENG	2002-05-30	farhana@gmail.com	105
6	Sakib Hossain	EEE	2001-09-14	sakib@gmail.com	106
7	Mim Akter	CSE	2002-12-02	mim@gmail.com	107
8	Arif Rahman	BBA	2001-04-21	arif@gmail.com	108
9	Shanto Das	CIVIL	2002-08-16	shanto@gmail.com	109
10	Lamia Chowdhury	CSE	2001-06-10	lamia@gmail.com	110

8. DELETE the student record with RollNo = 2:

```
1 • USE C243217_A00;
2
3 • DELETE FROM Dummy_students WHERE id=2;
4
```

SQL File 4* SQL File 5* SQL File 6* SQL File 7* SQL File 8*

```
1 -- SELECT * FROM students;
2 -- SELECT * FROM courses;
3 • SELECT * FROM Dummy_students;
4 -- SELECT * FROM Dummy_courses;
5
```

Result Grid Filter Rows: Edit: Export/Import:

id	name	dept	dob	email	Enrolled_CID
1	Rahim Uddin	CSE	2002-03-12	rahim@gmail.com	101
3	Nusrat Jahan	BBA	2002-01-18	nusrat@gmail.com	103
4	Tanvir Hasan	CSE	2001-11-09	tanvir@gmail.com	104
5	Farhana Akter	ENG	2002-05-30	farhana@gmail.com	105
6	Sakib Hossain	EEE	2001-09-14	sakib@gmail.com	106
7	Mim Akter	CSE	2002-12-02	mim@gmail.com	107
8	Arif Rahman	BBA	2001-04-21	arif@gmail.com	108
9	Shanto Das	CIVIL	2002-08-16	shanto@gmail.com	109
10	Lamia Chowdhury	CSE	2001-06-10	lamia@gmail.com	110

9. TRUNCATE the Courses table:

```
1 • USE C243217_A00;
2
3 • TRUNCATE TABLE Dummy_courses;
```

SQL File 4* SQL File 5* SQL File 6*

```
1 -- SELECT * FROM students;
2 -- SELECT * FROM courses;
3 -- SELECT * FROM Dummy_students;
4 • SELECT * FROM Dummy_courses;
5
```

Result Grid Filter Rows: Edit:

courseID	Course_Title	credits
NULL	NULL	NULL

10. ALTER the Students table to DROP the Email column you created earlier:

```
SQL File 4*  SQL File 5*  SQL File 6*  SQL File 7*  SQL File 8*  SQL F
Limit to 1000 rows
1 • USE C243217_A00;
2
3 • ALTER TABLE Dummy_students DROP COLUMN email;
```

```
SQL File 4*  SQL File 5*  SQL File 6*  SQL File 7*
Limit to 1000
1 -- SELECT * FROM students;
2 -- SELECT * FROM courses;
3 • SELECT * FROM Dummy_students;
4 -- SELECT * FROM Dummy_courses;
5
```

Result Grid | Filter Rows: | Edit:

	id	name	dept	dob	Enrolled_CID
▶	1	Rahim Uddin	CSE	2002-03-12	101
	3	Nusrat Jahan	BBA	2002-01-18	103
	4	Tanvir Hasan	CSE	2001-11-09	104
	5	Farhana Akter	ENG	2002-05-30	105
	6	Sakib Hossain	EEE	2001-09-14	106
	7	Mim Akter	CSE	2002-12-02	107
	8	Arif Rahman	BBA	2001-04-21	108
	9	Shanto Das	CIVIL	2002-08-16	109
	10	Lamia Chowdhury	CSE	2001-06-10	110
*	NULL	NULL	NULL	NULL	NULL

11. DROP the entire Students table:

```
SQL File 4*  SQL File 5*  SQL File 6*  SQL F
Limit to
1 • USE C243217_A00;
2
3 • DROP TABLE Dummy_students;
```

```
SQL File 4*  SQL File 5*  SQL File 6*  S
1 -- SELECT * FROM students;
2 -- SELECT * FROM courses;
3 • SELECT * FROM Dummy_students;
4 -- SELECT * FROM Dummy_courses;
5
```